

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 1: Mass Definitions & Structural Limits

Syllabus Source: CAE Oxford (pp. 1-22) | Pass Target: >= 90% AviationExam

1. THE CORE MASS DEFINITIONS HIERARCHY

In aviation, aircraft mass is divided into precise regulatory categories. Understanding the sequence of additions is vital:

Mass Term	Definition & Constituent Components	Master Equation & Significance
Basic Empty Mass (BEM)	Airframe structure, engines, unusable fuel, full engine oil, hydraulic fluid and permanent standard equipment.	Starting point of aircraft mass; manufacturer weighed.
Dry Operating Mass (DOM)	BEM + Flight Crew & Cabin Crew (+ their baggage) + Catering / Pantry + Removable emergency equipment.	DOM = BEM + Crew + Pantry. Mass of aircraft ready to fly without payload or fuel.
Operating Mass (OM)	DOM + Total Take-Off Fuel.	OM = DOM + Fuel. Does NOT include any passengers or cargo.
Zero Fuel Mass (ZFM)	DOM + Traffic Load (Payload: passengers + baggage + cargo).	ZFM = DOM + Traffic Load. The entire mass of aircraft excluding fuel!
Take-Off Mass (TOM)	ZFM + Total Take-Off Fuel, OR: DOM + Traffic Load + Take-Off Fuel.	TOM = ZFM + Fuel. Mass at brake release on the runway.
Landing Mass (LM)	TOM minus Trip Fuel consumed in flight.	LM = TOM - Trip Fuel.
Ramp / Taxi Mass	TOM + Taxi Fuel (allowance burned during taxi to runway).	Taxi Mass = TOM + Taxi Fuel (max allowable mass at gate).

2. PAYLOAD CONCEPTS: TRAFFIC LOAD VS USEFUL LOAD

- Traffic Load (Payload): The total mass of revenue-earning items: Passengers + Baggage + Cargo + Mail. Formula: Traffic Load = ZFM - DOM.
- Useful Load: The total mass of everything that can be loaded into an empty operating aircraft: Traffic Load + Usable Fuel. Formula: Useful Load = TOM - DOM.

3. STRUCTURAL LIMITS VS PERFORMANCE LIMITS

An aircraft is governed by two distinct sets of mass limits:

- Structural Limits (Maximum Certified): Permanent physical limits established by the manufacturer (MTOM, MZFM, MLM, MTXM). Exceeding these limits risks structural airframe failure (e.g. wing root bending overload at MZFM).
- Performance Limits (Variable / Regulated): Limits calculated for specific runway and environmental conditions (Runway Length Limited TOM, Climb Gradient Limited TOM, Obstacle Limited TOM).
- Golden Rule of Mass Planning: Actual TOM MUST NOT EXCEED the LOWEST of: (1) Structural MTOM, (2) Regulated Performance MTOM, (3) MZFM + Fuel, (4) MLM + Trip Fuel!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Maximum Allowable Take-Off Mass (MTOM)

SCENARIO (Typical AviationExam Drill): Given the following aircraft data: - Dry Operating Mass (DOM): 35,000 kg - Maximum Structural Take-Off Mass (MTOM): 65,000 kg - Maximum Structural Landing Mass (MLM): 56,000 kg - Maximum Structural Zero Fuel Mass (MZFM): 51,000 kg - Performance Limited TOM for Runway 09: 63,500 kg - Planned Fuel on board at Take-Off: 11,000 kg - Estimated Trip Fuel burn to destination: 4,000 kg

QUESTION: What is the MAXIMUM ALLOWABLE TAKE-OFF MASS that respects all limits? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check the Direct Take-Off Mass Limits: - Structural MTOM = 65,000 kg - Performance MTOM = 63,500 kg -> (This restricts us to 63,500 kg max directly). Step 2: Check the Zero Fuel Mass Limit constraint on TOM: - Max TOM = MZFM + Take-Off Fuel = 51,000 + 11,000 = 62,000 kg! - (If we took off at 63,500 kg with 11,000 kg of fuel, our ZFM would be 52,500 kg, which EXCEEDS the structural MZFM of 51,000 kg!). Step 3: Check the Landing Mass Limit constraint on TOM: - Max TOM = MLM + Trip Fuel = 56,000 + 4,000 = 60,000 kg! - (If we took off heavier than 60,000 kg, after burning 4,000 kg of trip fuel we would land above the 56,000 kg landing limit!). Step 4: Compare all 3 constrained Take-Off Masses: - Direct Take-off Limit: 63,500 kg - ZFM-Constrained Limit: 62,000 kg - Landing-Constrained Limit: 60,000 kg FINAL ANSWER: The Maximum Allowable Take-Off Mass is 60,000 kg! (Limited by Landing Mass).

[!] EXAM TRAP (AVIATIONEXAM) - The Wing-Root Bending / MZFM Exam Trap

- Why does MZFM exist? All payload is carried inside the fuselage, while fuel is carried inside the wings. In flight, wing fuel provides an upward relieving moment. When fuel burns down to zero, the entire heavy fuselage load is supported exclusively by the wing roots! Exceeding MZFM breaks the wing roots, even if you are well below MTOM!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the official definition of DRY OPERATING MASS (DOM)?**

[A] The mass of the airplane empty of all fuel and oil [B] Basic Empty Mass (BEM) PLUS crew, crew baggage, catering/pantry, and removable emergency equipment [C] Zero Fuel Mass minus payload [D] Maximum Take-Off Mass minus fuel

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. DOM is the total mass of the aircraft ready for a specific type of operation, including all crew, their personal baggage, and catering, but excluding all usable fuel and traffic load (payload).

Q2: If an aircraft's Maximum Zero Fuel Mass (MZFM) is 45,000 kg, Dry Operating Mass is 28,000 kg, and planned

[A] 25,000 kg [B] 17,000 kg [C] 9,000 kg [D] 12,000 kg

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Traffic Load = ZFM - DOM. The maximum possible ZFM is MZFM (45,000 kg). Therefore: Max Traffic Load = 45,000 - 28,000 = 17,000 kg.

Q3: Which component is included in the BASIC EMPTY MASS (BEM) of an aircraft?

[A] Flight crew baggage [B] Usable fuel in the wing tanks [C] Unusable fuel, engine oil, hydraulic fluid, and standard permanent equipment [D] Passenger catering supplies

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Basic Empty Mass includes the complete structural airframe, engines, standard fixed equipment, plus unusable fuel and full operating fluids (engine oil and hydraulic systems). Crew and catering are added to BEM to form DOM.

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 2: Center of Gravity Theory, Datums & Law of the Lever

Syllabus Source: CAE Oxford (pp. 23-42) | Pass Target: >= 90% AviationExam

1. CENTER OF GRAVITY (CG) PHYSICAL CONCEPT

The Center of Gravity (CG) is the single imaginary point through which the resultant force of gravity acts on the entire aircraft. If suspended by a cable attached precisely to its CG, the aircraft would remain in perfect horizontal equilibrium in any attitude.

2. THE DATUM & ARM SIGN CONVENTIONS

To calculate where the CG is located, all physical positions are measured from an arbitrary reference plane called the DATUM:

- The Datum: A vertical reference plane perpendicular to the longitudinal axis, chosen by the aircraft designer. Common locations: (1) Aircraft nose tip, (2) Engine firewall, (3) Wing leading edge.
- The Arm (Lever Arm): The horizontal distance from the datum to the center of gravity of an individual item (passenger seat row, cargo compartment, fuel tank).
- Sign Convention: By universal convention: Distance BEHIND (aft of) the datum is POSITIVE (+). Distance IN FRONT of (forward of) the datum is NEGATIVE (-).
- The Moment Equation: $\text{Moment} = \text{Mass} \times \text{Arm}$. Expressed in kg.m, kg.mm, or lb.in.

3. THE LAW OF THE LEVER & CENTER OF GRAVITY FORMULA

The total moment of an aircraft about the datum is equal to the algebraic sum of the individual moments of all items loaded onto it:

- Master CG Equation: $\text{Aircraft CG Location} = (\text{Total Moment}) / (\text{Total Mass}) = \text{Sum}(\text{Mass}_i \times \text{Arm}_i) / \text{Sum}(\text{Mass}_i)$.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Aircraft Center of Gravity from Loading Table

SCENARIO (Classic EASA Exam Question): An aircraft has the following loading schedule with the Datum located at the Engine Firewall: - Basic Empty Mass: 1,200 kg with an Arm of +1.50 m - Pilot and Copilot: 160 kg with an Arm of +1.10 m - Rear Passengers: 150 kg with an Arm of +2.20 m - Baggage in Nose Compartment: 40 kg with an Arm of -0.80 m (AHEAD of firewall!) - Fuel (in wings): 180 kg with an Arm of +1.60 m

QUESTION: Calculate the exact position of the aircraft's Center of Gravity relative to the Datum. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Build the Mass x Arm = Moment Table:

Item	Mass (kg)	Arm (m)	Moment (kg.m)
Item 1 (BEM)	1,200	+1.50	+1,800.0
Item 2 (Front)	160	+1.10	+176.0
Item 3 (Rear)	150	+2.20	+330.0
Item 4 (Nose Bag)	40	-0.80	-32.0
Item 5 (Fuel)	180	+1.60	+288.0

[NOTE: NEGATIVE MOMENT!] - Item 5 (Fuel): 180 kg x (+1.60 m) = +288.0 kg.m

Step 2: Sum the Total Mass: - Total Mass = 1,200 + 160 + 150 + 40 + 180 = 1,730 kg. Step 3: Sum the Algebraic Moments (be careful with the negative sign!): - Total Moment = 1,800.0 + 176.0 + 330.0 - 32.0 + 288.0 = +2,562.0 kg.m. Step 4: Divide Total Moment by Total Mass: - CG = Total Moment / Total Mass = +2,562.0 kg.m / 1,730 kg = +1.4809 m. FINAL ANSWER: The Center of Gravity is located at +1.48 m (1.48 meters aft of the firewall).

[!] EXAM TRAP (AVIATIONEXAM) - The Forward Datum / Negative Arm Trap

- If an item is placed IN FRONT of the datum, its arm is NEGATIVE (-). - Its moment is therefore NEGATIVE! (Mass x (-Arm) = -Moment). - In the total moment sum, you MUST SUBTRACT this negative moment, but you MUST STILL ADD its mass to the total mass!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the definition of the 'ARM' of a load in mass and balance calculations?**

[A] The vertical height of the item above the cabin floor [B] The horizontal distance from the reference DATUM to the center of gravity of the item [C] The length of the fuselage [D] The structural thickness of the wing spar

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The arm is the physical distance along the longitudinal axis from the established reference datum to the center of gravity of the specific item.

Q2: If an aircraft has a total mass of 2,000 kg and a total moment of 6,000 kg.m about the datum, where is the CG?

[A] 4.0 m aft of datum [B] 3.0 m aft of datum [C] 0.33 m aft of datum [D] At the datum

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. CG = Total Moment / Total Mass = 6,000 kg.m / 2,000 kg = 3.0 meters aft of the datum.

Q3: If a designer chooses the aircraft nose as the Datum, what sign will all arms of items loaded inside the aircraft have?

[A] Negative [B] Positive (+), because all loaded items are located aft of the nose tip [C] Zero [D] Variable depending on flight speed

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. When the datum is placed at the extreme front tip (nose), all positions in the cabin, wings, and empennage lie aft of the datum, meaning all arms and moments are positive (+), eliminating negative numbers.

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 3: Calculation of CG & % MAC (Mean Aerodynamic Chord)

Syllabus Source: CAE Oxford (pp. 43-66) | Pass Target: >= 90% AviationExam

1. THE MEAN AERODYNAMIC CHORD (MAC) CONCEPT

Because modern aircraft have swept, tapered wings, the aerodynamic chord varies along the wingspan. The Mean Aerodynamic Chord (MAC / $c_{\bar{}}$) is the chord of an equivalent rectangular wing that has the identical pitching moment and aerodynamic characteristics as the actual wing!

- LEMAC: Leading Edge of the Mean Aerodynamic Chord. The distance from the reference Datum to the front edge of the MAC.
- TEMAC: Trailing Edge of the Mean Aerodynamic Chord. LEMAC + Length of MAC = TEMAC.
- Why Express CG as % MAC?: Expressing CG as a percentage of MAC allows pilots to understand the aerodynamic stability and trim of an aircraft immediately, regardless of whether the aircraft is a small trainer or a heavy Boeing 777!

2. THE MASTER CONVERSION FORMULAS: DISTANCE <-> % MAC

- Formula 1: Converting CG Distance into % MAC: $\% \text{ MAC} = ((\text{CG Distance from Datum} - \text{LEMAC}) / \text{Length of MAC}) \times 100$
- Formula 2: Converting % MAC into CG Distance from Datum: $\text{CG Distance from Datum} = \text{LEMAC} + (\% \text{ MAC} / 100) \times \text{Length of MAC}$

3. STABILITY & CONTROLLABILITY TRADE-OFF AT THE LIMITS

CG Position Regime	Aerodynamic Pitch Stability & Stick Forces	Flight Control Authority & Fuel Performance
CG Position Regime	Aerodynamic Pitch Stability & Stick Forces	Flight Control Authority & Fuel Performance
Forward CG Limit	MAXIMUM static longitudinal stability. Very heavy stick forces (high stick force per g).	Landing flare authority severely limited! High downward tailplane force -> high induced trim drag and higher fuel burn.
Aft CG Limit	MINIMUM static stability. Light, sensitive stick forces. Danger of over-controlling.	Lower tailplane downforce -> LOWER TRIM DRAG, lower fuel burn. Must NOT exceed Neutral Point (neutral/unstable!).

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Converting CG Position to % MAC and Vice Versa

SCENARIO 1 (Finding % MAC): Given the following aircraft technical data: - LEMAC is located at: 14.0 meters aft of datum - Length of the MAC: 4.0 meters - Actual aircraft CG calculated from loading sheet: 15.2 meters aft of datum QUESTION: What is the aircraft's Center of Gravity expressed as a percentage of MAC (% MAC)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the distance from LEMAC to the CG: - Distance from LEMAC = CG - LEMAC = 15.2 m - 14.0 m = 1.2 meters. Step 2: Divide this distance by the total length of the MAC: - Fraction of MAC = 1.2 m / 4.0 m = 0.30. Step 3: Multiply by 100 to get a percentage: - % MAC = 0.30 x 100 = 30.0% MAC. FINAL ANSWER: CG = 30.0% MAC. ----- SCENARIO

2 (Finding Physical CG Distance from % MAC): QUESTION: If the certified forward limit is 18% MAC, at what distance from the datum is the forward limit located? SOLUTION PASO A PASO: Step 1: Convert percentage to decimal: 18% = 0.18. Step 2: Multiply decimal by MAC length: 0.18 x 4.0 m = 0.72 meters aft of LEMAC. Step 3: Add this distance to LEMAC: 14.0 m + 0.72 m = 14.72 meters. FINAL ANSWER: Forward CG Limit is at 14.72 m aft of Datum.

[!] EXAM TRAP (AVIATIONEXAM) - TEMAC and MAC Length Trap

- If an exam question gives you LEMAC and TEMAC instead of MAC length: - Formula: Length of MAC = TEMAC - LEMAC! - E.g. If LEMAC = 500 inches and TEMAC = 700 inches, the MAC length is 200 inches.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If LEMAC is at station 20 m and TEMAC is at station 25 m, what is the length of the Mean Aerodynamic Chord?

[A] 45 m [B] 5 m [C] 22.5 m [D] 10 m

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. MAC Length = TEMAC - LEMAC = 25 m - 20 m = 5 meters.

Q2: An aircraft has LEMAC at 1,000 inches and MAC length of 150 inches. If the CG is at 26% MAC, what is the CG position?

[A] 1,039 inches [B] 1,026 inches [C] 1,150 inches [D] 1,015 inches

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. CG = LEMAC + (% MAC / 100 x MAC) = 1,000 + (0.26 x 150) = 1,000 + 39 = 1,039 inches.

Q3: Why does flying with a Center of Gravity at the AFT limit result in lower fuel consumption than at the FORWARD limit?

[A] The engines produce more thrust [B] Less tailplane download is required to balance the aircraft, which reduces total lift required and lowers induced trim drag [C] The aircraft flies at lower TAS [D] Air density increases

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. At an aft CG, the nose-down pitching moment is small, requiring a smaller downward balancing force from the horizontal stabilizer. Lower total lift required means lower induced trim drag, saving fuel.

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 4: Mass Shifts, Load Changes & Ballast Calculations

Syllabus Source: CAE Oxford (pp. 67-88) | Pass Target: $\geq 90\%$ AviationExam

1. THE FUNDAMENTAL MASS SHIFT FORMULA

When an item of mass already on board is moved from one compartment to another, the aircraft's Total Mass DOES NOT CHANGE, but the Center of Gravity shifts in the direction of the movement!

- The Universal Mass Shift Formula: $(\text{Mass Shifted (m)}) / (\text{Total Aircraft Mass (M)}) = (\text{Change in CG (Delta CG)}) / (\text{Distance Shifted (d)})$
- Solving for Delta CG: $\text{Delta CG} = (m \times d) / M$
- Solving for Mass to Shift (m): $m = (M \times \text{Delta CG}) / d$
- Solving for Distance to Shift (d): $d = (M \times \text{Delta CG}) / m$

2. ADDING OR REMOVING MASS

When mass is added or removed, Total Mass CHANGES. Use the Moment equation:

- New CG Equation: $\text{New CG} = (\text{Old Total Moment} \pm \text{Added/Removed Moment}) / (\text{Old Total Mass} \pm \text{Added/Removed Mass})$

3. BALLAST CALCULATIONS (CORRECTING OUT-OF-LIMITS CG)

If an empty or lightly loaded ferry flight has a CG aft of the aft limit, ballast mass must be loaded into a forward cargo bay to bring the CG exactly onto the limit:

- Ballast Formula: $\text{Ballast Mass (m}_b) = (\text{Total Mass} \times (\text{Current CG} - \text{Desired Limit CG})) / (\text{Desired Limit CG} - \text{Ballast Arm})$

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Cargo Shift to Bring CG to Forward Limit

SCENARIO (Extremely Common AviationExam Question): An aircraft has: - Total Mass (M):

60,000 kg - Current CG position: 22.0 meters aft of datum - Certified Aft CG Limit:

21.6 meters aft of datum (The aircraft is 0.4 m OUT OF LIMITS!) - Forward Cargo Hold

is at: 10.0 meters aft of datum - Aft Cargo Hold is at: 30.0 meters aft of

datum QUESTION: How much cargo mass must be shifted from the Aft Cargo Hold to the

Forward Cargo Hold to bring the CG to the Aft Limit of 21.6 m? SOLUTION PASO A PASO

(FOR DUMMIES): Step 1: Identify all known variables: - Total Mass (M) = 60,000 kg

- Desired change in CG (Delta CG) = Current CG - Target CG = 22.0 m - 21.6 m = 0.40

meters. - Distance of shift (d) = Distance from Aft Hold to Fwd Hold = 30.0 m - 10.0

m = 20.0 meters. Step 2: Recall the Master Shift Formula: - $m / M = \text{Delta CG} / d$

-> $m = (M \times \text{Delta CG}) / d$ Step 3: Plug in the numbers and calculate: - $m = (60,000$

kg $\times 0.40 \text{ m}) / 20.0 \text{ m}$ - $m = 24,000 / 20.0 = 1,200 \text{ kg}$. FINAL ANSWER: You must shift

exactly 1,200 kg of cargo from the Aft Hold to the Forward Hold!

[!] EXAM TRAP (AVIATIONEXAM) - The Direction of Shift Trap

- Shifting mass FORWARD moves the aircraft CG FORWARD (decreases CG distance from nose).
- Shifting mass AFT moves the aircraft CG REARWARD (increases CG distance from nose).
- Distance of shift (d) is ALWAYS the positive physical distance between the two compartments: $d = |\text{Hold_A} - \text{Hold_B}|$.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: An aircraft has a gross mass of 5,000 kg and CG at station 120 inches. If 100 kg of baggage is shifted from

[A] 118 inches [B] 115 inches [C] 122 inches [D] 110 inches

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. Shift distance $d = 150 - 50 = 100$ inches (forward). $\Delta \text{CG} = (m \times d) / M = (100 \text{ kg} \times 100 \text{ in}) / 5,000 \text{ kg} = 10,000 / 5,000 = 2.0$ inches forward. New CG = $120 - 2.0 = 118$ inches.

Q2: In mass shift calculations, what happens to the total mass of the aircraft?

[A] It increases by the amount of mass moved [B] It decreases [C] It remains EXACTLY CONSTANT [D] It shifts by Delta CG

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Moving an object from one cabin location to another redistributes internal forces but does not add or subtract mass. Total aircraft mass M remains constant.

Q3: If 500 kg of cargo is REMOVED from a cargo bay located forward of the aircraft's current CG, what will hap

[A] It will move forward [B] It will move REARWARD (aft) [C] It will remain stationary [D] It becomes negative

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Removing mass from in front of the CG takes away forward weight, causing the remaining balance to tilt backwards (CG moves aft).

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 5: Fuel Planning & Non-Linear CG Movement

Syllabus Source: CAE Oxford (pp. 89-106) | Pass Target: >= 90% AviationExam

1. FUEL TANKS GEOMETRY & CG SHIFT IN FLIGHT

Unlike passenger mass or cargo which remain stationary in flight, fuel mass decreases continuously throughout the trip. Because fuel tanks have complex 3D shapes across swept wings, fuel consumption causes non-linear Center of Gravity movement:

- **Swept-Wing Fuel Tank Arm:** As wings sweep backwards, wingtip tanks are located significantly further AFT than the wing root center tank. Burning fuel from different tanks dramatically moves the aircraft CG forward or aft!
- **Wing Bending Relief & Standard Burn Sequence:** To minimize structural wing root bending moments, commercial jets burn fuel in a strict sequence: CENTER TANK FIRST, WING TANKS LAST! Burning the fuselage center tank first relieves wing-root stress because fuel weight inside the wings counteracts aerodynamic upward wing lift.
- **Unusable Fuel (Trapped Fuel):** Fuel remaining in pipework, pumps, and tank bottoms that cannot be drained into engines during flight. **REGULATION:** Unusable fuel is legally part of BASIC EMPTY MASS (BEM)! It is NEVER counted as usable trip fuel.

2. IN-FLIGHT FUEL JETTISON (FUEL DUMPING)

If a heavy aircraft experiences an emergency immediately after take-off, it may be above its Maximum Structural Landing Mass (MLM):

- **Fuel Jettison System:** Dumps fuel overboard through wingtip nozzles to reduce aircraft mass down to MLM. Jettison valves close automatically to preserve reserve fuel (minimum holding fuel to land).
- **CG Impact of Jettisoning:** Because jettisoning typically empties center tanks and outer wing tanks rapidly, the flight crew must monitor that the CG remains strictly within the certified in-flight envelope during dumping.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Tracking CG Movement from Take-Off to Landing

SCENARIO (Typical Exam Question): An aircraft departs with: - Take-Off Mass (TOM):

50,000 kg - Take-Off CG: 18.00 meters aft of datum - Total Take-Off Moment = 50,000 x

18.00 = 900,000 kg.m - Trip Fuel to be burned in flight: 8,000 kg - The average Arm of the burned fuel is: 21.00 meters aft of datum (located AFT of the CG!) QUESTION:

Calculate the aircraft's Landing Mass (LM) and the new Landing CG position. SOLUTION

PASO A PASO (FOR DUMMIES): Step 1: Calculate Landing Mass (LM): - LM = TOM - Trip

Fuel = 50,000 kg - 8,000 kg = 42,000 kg. Step 2: Calculate the Moment of the Burned

Fuel: - Fuel Moment = Fuel Mass x Fuel Arm = 8,000 kg x 21.00 m = 168,000

kg.m. Step 3: Subtract the burned fuel moment from the Take-Off Moment: - Landing

Moment = Take-Off Moment - Fuel Moment - Landing Moment = 900,000 kg.m - 168,000

kg.m = 732,000 kg.m. Step 4: Divide Landing Moment by Landing Mass to find Landing

CG: - Landing CG = Landing Moment / Landing Mass - Landing CG = 732,000 kg.m /

42,000 kg = 17.4285 m -> 17.43 meters aft of datum! PHYSICAL EXPLANATION FOR

DUMMIES: Because the burned fuel was located at 21 m (AFT of the 18 m CG), burning it removed weight from the back of the airplane, causing the nose to tilt down and

shifting the CG FORWARD from 18.00 m to 17.43 m!

[!] EXAM TRAP (AVIATIONEXAM) - The Non-Linear Fuel Curve Trap

- Center tanks burn first -> CG typically moves AFT initially. - Wing tanks burn

second -> Swept-wing fuel burn moves CG back FORWARD. - Exam Trap: You CANNOT assume

that if Take-off CG and Landing CG are both within limits, the aircraft was in limits

throughout the entire cruise! You must check the intermediate burn curve peak!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under EASA AIR-OPS, which category does 'UNUSABLE FUEL' belong to?**

[A] Trip Fuel [B] Contingency Fuel [C] BASIC EMPTY MASS (BEM) [D] Traffic Load

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Unusable fuel is the fuel trapped in pipes and tank sumps that cannot reach the engines. It is a permanent fluid load counted as part of the aircraft's Basic Empty Mass.

Q2: Why is fuel in commercial jet aircraft burned from the center fuselage tank before the wing tanks?

[A] Center tank fuel spoils faster [B] Keeping fuel inside the wing tanks as long as possible provides upward inertia and wing bending relief against aerodynamic lift [C] The center pump has higher pressure [D] Wing tanks cannot be pressurized

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Wing fuel weight acts downward along the wingspan, opposing the upward aerodynamic lift, reducing wing-root bending stress. Emptying the center tank first maximizes this structural relief.

Q3: If fuel burned during cruise has an arm located AFT of the current Center of Gravity, what happens to the CG?

[A] The CG moves FORWARD [B] The CG moves rearward [C] The CG stays frozen [D] The CG becomes negative

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. Removing mass from behind the pivot point causes the remaining balance to tilt forward, shifting the CG forward.

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 6: Standard Masses, Floor Limits & Cargo Restraints

Syllabus Source: CAE Oxford (pp. 107-124) | Pass Target: >= 90% AviationExam

1. EASA AIR-OPS STANDARD PASSENGER MASSES

Commercial operators may use EASA standard passenger masses instead of weighing every individual passenger at the gate:

Passenger Category	All Flights (>= 30 Passenger Seats)	Holiday Charter (>= 30 Seats)	Aircraft with 1 to 19 Seats
Passenger Category	All Flights (>= 30 Passenger Seats)	Holiday Charter (>= 30 Seats)	Aircraft with 1 to 19 Seats
Male Passenger	88 kg (including 6 kg hand luggage)	83 kg	96 kg (Male)
Female Passenger	70 kg (including 6 kg hand luggage)	69 kg	78 kg (Female)
All Adults (No Gender Split)	84 kg	76 kg	88 kg
Children (2 to 11 years)	35 kg	35 kg	35 kg
Infants (Under 2 years)	0 kg (if held on adult's lap)	0 kg	0 kg (Held on lap)

2. CARGO COMPARTMENT LIMITATIONS: FLOOR & RUNNING LOADS

Cargo holds are protected against structural collapse by two strict engineering limitations:

- Area Load Limit (Floor Intensity): Maximum allowable mass per unit surface area: kg/m^2 or lb/ft^2. Exceeding this punches through the floor panel! Formula: Load Intensity = Mass / Contact Area <= Floor Limit.
- Running Load Limit (Linear Load Limit): Maximum allowable mass per unit length of fuselage: kg/m or lb/in. Exceeding this breaks the transverse fuselage frames! Formula: Linear Load = Mass / Length <= Running Limit.
- Spreader Plates (Dunnage): If a dense, heavy box exceeds the floor load limit, placing rigid wooden or composite spreader plates beneath it increases the contact area, distributing the weight safely across multiple floor beams!

3. CARGO RESTRAINT & TIE-DOWN LOAD FACTORS

To prevent cargo breaking free during severe turbulence or emergency landing, lashings must restrain against design load factors (EASA CS-25): Forward: 9.0 g; Upward: 1.5 g; Sideward: 3.0 g; Aft: 1.5 g.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Floor Load Limits & Spreader Sizing

SCENARIO (Classic EASA Calculation Drill): A cargo compartment has the following certified structural limits: - Maximum Floor Load Limit: 600 kg/m² - Maximum Running Load Limit: 400 kg/m A piece of industrial machinery has a mass of 1,200 kg. Its base dimensions are: Length = 2.0 meters, Width = 0.8 meters. QUESTION: Does this cargo exceed the Floor Load Limit or the Running Load Limit? If it exceeds, what is the MINIMUM contact area required using a spreader plate? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check the Running Load Limit: - Actual Linear Load = Mass / Length = 1,200 kg / 2.0 m = 600 kg/m. - Certified Running Limit = 400 kg/m. - VERDICT: 600 kg/m > 400 kg/m -> EXCEEDS RUNNING LOAD LIMIT by 200 kg/m! Step 2: Check the Floor Load (Area) Limit: - Contact Area = Length x Width = 2.0 m x 0.8 m = 1.6 m². - Actual Area Load Intensity = Mass / Area = 1,200 kg / 1.6 m² = 750 kg/m². - Certified Floor Limit = 600 kg/m². - VERDICT: 750 kg/m² > 600 kg/m² -> EXCEEDS FLOOR LOAD LIMIT by 150 kg/m²! Step 3: Calculate Minimum Spreader Plate Area required: - Area_min = Mass / Max Floor Limit = 1,200 kg / 600 kg/m² = 2.0 m². FINAL ANSWER: Both limits are exceeded! A spreader plate of AT LEAST 2.0 m² contact area and at least 3.0 m length (1,200/400) is mandatory.

[!] EXAM TRAP (AVIATIONEXAM) - Infant Mass & Hand Luggage Rules

- Infant mass (under 2 years) is 0 kg ONLY if held on an adult's lap. If occupying an individual seat, child mass (35 kg) must be applied! - Hand luggage (6 kg) is ALREADY INCLUDED in the standard passenger mass (88 kg male = 82 kg person + 6 kg bag). Do not add 6 kg twice!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA AIR-OPS on an aircraft with 30 or more passenger seats, what is the standard mass for an adult male passenger?

[A] 84 kg [B] 88 kg (including 6 kg hand luggage allowance) [C] 70 kg [D] 96 kg

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The EASA standard male passenger mass for aircraft with 30+ seats is 88 kg (82 kg body mass + 6 kg hand luggage).

Q2: What is the primary operational purpose of placing 'SPREADER PLATES' (dunnage) under dense cargo in a cargo hold?

[A] To increase aircraft payload [B] To increase the effective contact surface area, reducing floor load intensity (kg/m²) below the certified structural floor limit [C] To prevent fire [D] To move the CG forward

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Spreader plates distribute the concentrated downward weight of heavy, compact cargo across a wider surface area, ensuring local pressure does not exceed the floor structural limit.

Q3: Under CS-25 regulations, what is the forward ultimate inertia load factor that cargo tie-downs and restraint devices must withstand?

[A] 1.5 g [B] 3.0 g [C] 9.0 g forward [D] 20.0 g

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. To protect occupants in an emergency landing, cargo restraints must hold cargo in place against a 9.0 g forward deceleration load factor.

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 7: Load & Trim Sheet Execution & STAB TRIM Setting

Syllabus Source: CAE Oxford (pp. 125-144) | Pass Target: $\geq 90\%$ AviationExam

1. THE ARCHITECTURE OF THE LOAD AND TRIM SHEET

The Load and Trim Sheet is the official legal document required before every commercial departure. It fulfills two vital safety functions:

- **Mass Check (The Upper Section):** Verifies that the aircraft does not exceed any of its 3 structural mass limits: Maximum Zero Fuel Mass (MZFM), Maximum Take-Off Mass (MTOM), and Maximum Landing Mass (MLM).
- **Balance & Trim Check (The Lower Section):** Graphically plots the Center of Gravity using an 'Index Grid' (trim drop lines) to verify that the CG is inside the flight envelope and to read the Take-Off Stabilizer Trim setting (STAB TRIM in units ANU / AND).

2. THE INDEX SYSTEM & REDUCTION FORMULA

Because raw moments on transport jets run into millions of kilogram-meters, calculations are simplified into small manageable 'Index Units':

- **The Index Formula:** $\text{Index} = ((\text{Mass} \times (\text{Arm} - \text{Reference Datum})) / \text{Divisor (C)}) + \text{Constant (K)}$
- **Divisor (C):** Typically 10,000 or 100,000. Reduces the large moment into a small 2 or 3 digit number.
- **Constant (K):** A positive baseline offset (e.g. +50 or +100) added so that all calculated indices remain strictly positive numbers, eliminating negative calculation errors!

3. TAKE-OFF STABILIZER TRIM (STAB TRIM SETTING)

Before take-off, the trimmable horizontal stabilizer (THS) must be set to a specific pitch trim angle corresponding to the Take-Off % MAC:

- **Aft CG -> Low Nose-Up Trim:** At an aft CG, the aircraft naturally pitches up easily. STAB TRIM is set to a low nose-up value (e.g. 3.0 units ANU).
- **Forward CG -> High Nose-Up Trim:** At a forward CG, heavy nose-down moments require a large nose-up stabilizer setting (e.g. 7.5 units ANU) to allow rotation at V_r without excessive pilot pull force!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Index & Reading Trim Sheet

SCENARIO (Standard EASA Exam Trim Sheet Problem): An airline uses the following Index

Reduction Formula for its twin-jet: $\text{Index} = [\text{Mass (kg)} \times (\text{Arm (m)} - 15.0 \text{ m}) / 1,000]$

+ 50.0 Given the following loading changes to the Dry Operating Index: - Dry Operating

Mass (DOM): 32,000 kg with an Initial Index of 52.0 units - Cargo loaded in

Compartment 1: 2,000 kg at Arm = 8.0 meters - Cargo loaded in Compartment 4: 1,500 kg

at Arm = 22.0 meters QUESTION: Calculate the New Zero Fuel Mass (ZFM) and the New Zero

Fuel Index! SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate New Zero Fuel

Mass: - $\text{ZFM} = \text{DOM} + \text{Compartment 1} + \text{Compartment 4}$ - $\text{ZFM} = 32,000 \text{ kg} + 2,000 \text{ kg} +$

1,500 kg = 35,500 kg. Step 2: Calculate Delta Index for Compartment 1 (Arm = 8.0 m):

- $\text{Delta Index}_1 = [\text{Mass} \times (\text{Arm} - 15.0)] / 1,000$ - $\text{Delta Index}_1 = [2,000 \times (8.0 -$

15.0)] / 1,000 = $[2,000 \times (-7.0)] / 1,000 = -14.0$ units! - (Notice it is negative

because the arm is forward of the 15.0 m reference!). Step 3: Calculate Delta Index

for Compartment 4 (Arm = 22.0 m): - $\text{Delta Index}_4 = [\text{Mass} \times (\text{Arm} - 15.0)] / 1,000$

- $\text{Delta Index}_4 = [1,500 \times (22.0 - 15.0)] / 1,000 = [1,500 \times (+7.0)] / 1,000 = +10.5$

units! Step 4: Sum the Initial Index with both Delta Indices to find New ZFM Index:

- $\text{New ZFM Index} = \text{Initial Index} + \text{Delta Index}_1 + \text{Delta Index}_4$ - $\text{New ZFM Index} =$

52.0 + (-14.0) + (+10.5) = 48.5 units! FINAL ANSWER: New ZFM = 35,500 kg; New ZFM

Index = 48.5 units. On the trim grid, enter at Index 48.5 to read % MAC.

[!] EXAM TRAP (AVIATIONEXAM) - STAB TRIM Setting Units Trap (ANU vs AND)

- ANU = Aircraft Nose Up (positive nose-up trim). - AND = Aircraft Nose Down. -

Forward CG ALWAYS requires MORE ANU (higher nose-up stabilizer trim) to assist elevator rotation at Vr!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary purpose of using 'INDEX UNITS' instead of raw moments in aircraft load and trim sheet**

[A] To convert metric units to imperial units [B] To reduce huge moment numbers (millions of kg.m) into manageable small numbers, preventing human calculation errors in time-critical airline turnarounds [C] To calculate cruise speed [D] To record passenger names

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Moments for transport airliners are enormous numbers. Dividing by a constant divisor (e.g. 10,000) and adding a base constant produces compact index values (e.g. 40-60) easily plotted on graphical grids.

Q2: When setting the Take-Off Stabilizer Trim (STAB TRIM) on a commercial transport jet, what setting is required

[A] Zero trim [B] A GREATER Aircraft Nose Up (ANU) trim setting, to counter heavy nose-down moments and ensure adequate elevator rotation authority at Vr [C] Full Aircraft Nose Down (AND) trim [D] Trim does not depend on CG

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. A forward CG produces a large nose-down moment. A higher ANU (nose-up) stabilizer setting is mandatory so the pilot can rotate the nose at Vr without exceeding control column pull limits.

Q3: On a load and trim sheet, if an aircraft's Take-Off Mass is plotted and falls OUTSIDE the certified envelope

[A] Increase engine thrust to TOGA [B] Departure is STRICTLY PROHIBITED until payload or fuel is redistributed or offloaded to bring the point within the certified envelope [C] Notify ATC on climb out [D] Fly at a lower altitude

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Flying outside the M&B envelope is illegal and causes loss of control or structural overload. The flight crew must re-trim by shifting cargo, moving passengers, or offloading fuel before departure.